

LAB 3. BASIC OBJECT-ORIENTED TECHNIQUES

IT3103E – 162065

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- Question: ***Is JAVA a Pass by Value or a Pass by Reference programming language?***

Java is a Pass by Value programming language.

- After the call of **swap(jungleDVD, cinderellaDVD)** why does the title of these two objects still remain?

We try to change the value of jungleDVD and cinderellaDVD for each other by swap() method but when we call swap(), it only changes the direction of the pointer from jungleDVD and cinderellaDVD to the local copies but not the object. So when we call the getter, the returned titles of them still remain the same.

- After the call of **changeTitle(jungleDVD, cinderellaDVD.getTitle())** why is the title of the JungleDVD changed?

We use changeTitle() to make jungleDVD hold the title of cinderellaDVD through its setter method. Inside the method, we modify the content of the object that jungleDVD refers. We force it to hold the title "Cinderella", it doesn't change the object the reference point to itself.

- **Suggestion:** Write a **toString()** method for the **DigitalVideoDisc** class. What should be the return type of this method?

String should be the return type

- Read a text/binary file to a **String** without using **StringBuffer** to concatenate String (only use **+** operator). Observe and capture your screen when you choose a very long file

```
1 package hust.soict.dsai.garbage;
2
3 import java.nio.file.Files;
4
5
6 public class GarbageCreator {
7     public static void main(String[] args) throws Exception {
8         String filename = "src/hust/soict/dsai/garbage/test.txt";
9         byte[] inputBytes = Files.readAllBytes(Paths.get(filename));
10
11         long startTime = System.currentTimeMillis();
12
13         String outputString = "";
14         for (byte b : inputBytes) {
15             outputString += (char) b;
16         }
17
18         long endTime = System.currentTimeMillis();
19         System.out.println(endTime - startTime);
20     }
21 }
22
```

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<terminated> GarbageCreator [Java Application] C:\Program Files\Java\jdk-1.8\bin\javaw.exe (Oct 2052970

- Improve the code using **StringBuffer**

```
1 package hust.soict.dsai.garbage;
2
3 import java.nio.file.Files;
4 import java.nio.file.Paths;
5
6 public class NoGarbage {
7     public static void main(String[] args) throws Exception {
8         String filename = "src/hust/soict/dsai/garbage/test.txt";
9         byte[] inputBytes = Files.readAllBytes(Paths.get(filename));
10
11         long startTime = System.currentTimeMillis();
12
13         StringBuilder outputStringBuilder = new StringBuilder();
14         for (byte b : inputBytes) {
15             outputStringBuilder.append((char) b);
16         }
17
18         long endTime = System.currentTimeMillis();
19         System.out.println(endTime - startTime);
20     }
21 }
22
```

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<terminated> NoGarbage [Java Application] C:\Program Files\Java\jdk-1.8\bin\javaw.exe (Oct 31, 2017)
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